

Detecting AI-Generated Text

A comparative evaluation of classical and Transformer-based classifiers

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The task and the data

Kaggle — AI vs Human Text · 487,235 essays

62.8%

human-written

37.2%

AI-generated

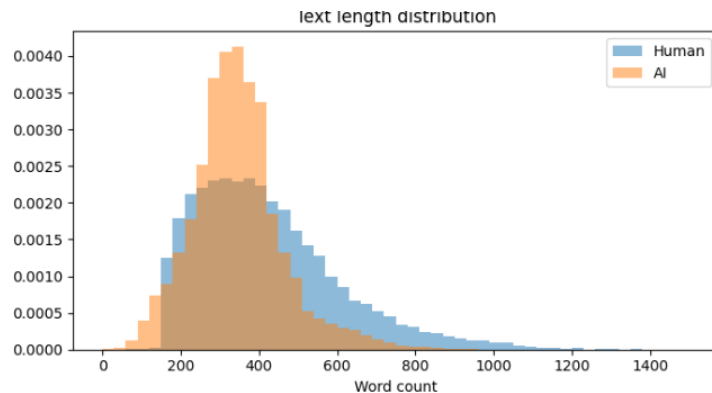
487,199

after filtering

Imbalance matters: a model that always predicts "human" scores 62.8% — so we report macro-F1, not accuracy alone.

Length differs by class: human median 390 words (sd 187) vs AI 337 words (sd 117). AI text is shorter and far more uniform.

Filtering: 36 junk texts under 20 words removed — all of them AI-labelled.



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Method

Two model families, one shared split

Classical

- TF-IDF, unigrams + bigrams
- Logistic Regression
- Multinomial Naive Bayes
- Linear SVM
- `class_weight='balanced'` (LR, SVM)

Transformer

- `distilbert-base-uncased`
- Hugging Face Trainer, Colab T4, fp16
- 40k stratified training subsample
- `max_length = 256` tokens
- 2 hyperparameter configurations

Same stratified 80/20 split, `random_state = 42`, reused across all four notebooks. Both families are scored on the identical 97,440 held-out essays — the comparison is between models, not data handling.

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Results

Shared test set, $n = 97,440$

Model	Accuracy	F1 (AI)	Macro-F1
Linear SVM + TF-IDF	0.9997	0.9996	0.9997
Logistic Regression + TF-IDF	0.9963	0.9950	0.9960
DistilBERT (fine-tuned)	0.9947	0.9929	0.9943
Multinomial Naive Bayes	0.9727	0.9626	0.9705

**Both linear models beat
the fine-tuned
Transformer**

Why did the simpler model win? Four non-exclusive explanations:

Linear separability

dominant signal is lexical — a sparse linear model is built for it

10× less data

390k training rows vs a 40k subsample

Truncation

256 tokens $\approx$ 190 words; TF-IDF saw whole documents

Possible leakage

near-duplicates may straddle the split

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What the models actually learned

The strongest "human" signals are artifacts of how the dataset was assembled

`teacher_name` `school_name`
`proper_name`

Anonymisation placeholders — present only in the human half of the corpus.

The human essays were redacted before release. The AI essays never were — there was no name to redact.

This has nothing to do with how the two kinds of text are written, and everything to do with how the dataset was built.

Also leaking the label:

Typos

driveless, wont — humans err, AI rarely does

Topic words

nasa, venus, car — prompt distribution, not style

Corrupted text

"TLE work", "iy Roiert Duffer" — encoding damage

The models may be detecting this dataset — not AI writing.

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Error analysis

Where each model fails — and where they overlap

31

SVM errors

518

DistilBERT errors

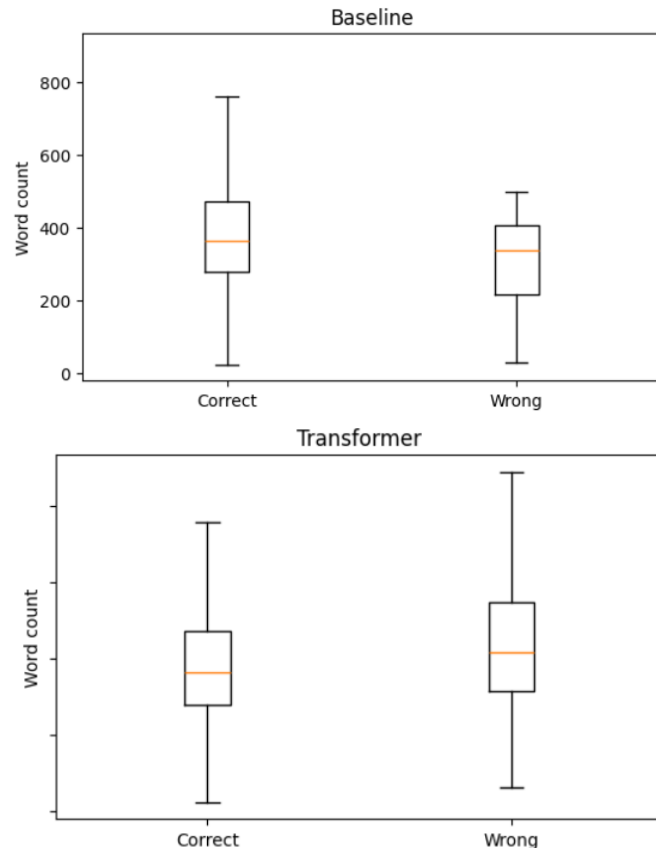
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shared by both

Truncation. DistilBERT's errors run longer than its correct predictions (median 414.5 vs 362 words); 83.8% fall on essays over 256 words, vs 67.7% for the SVM — which has no truncation limit to explain it.

Shared blind spot. AI text that is caught contains formulaic phrases 88.7% of the time; AI text that is missed, only 64.5%. Both models lean on the same surface cue.

Near-duplicate essays differing only by character corruption appear in the test set — the duplicate check is still open.



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Ethics and conclusion

Bias against non-native writers

Detectors are documented to misflag non-native English writing (Liang et al., 2023). Our AI-associated vocabulary sits in the same formal register.

Asymmetric error costs

A false accusation harms a real student; a missed detection does not. Both models optimised as if the two were equal.

Not deployable as-is

Performance rests partly on dataset artifacts and is untested outside this corpus.

Next step — the cheapest experiment is the most revealing: strip the anonymisation placeholders, retrain both models, and measure how much of 99.97% survives.